

P5 Quantum benchmark

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One of the reasons we chose a 1D anharmonic oscillator is because, we can obtain the exact quantum momentum distribution, by diagonalising the Hamiltonian for benchmark with relative ease. We will walk through the maths behind what the code `Exact_quartic.f90` implements. For Hamiltonian $H = H_0 + q^4/4$, where $H_0 = (p^2 + q^2)/2$ we wish to calculate the momentum distribution

$$n(p) = \frac{\langle p | e^{-\beta H} | p \rangle}{\text{tr}[e^{-\beta H}]} = \sum_j w_j |\langle p | \psi_j \rangle|^2 \quad (1)$$

where $w_j = e^{-\beta E_j} / \sum_k e^{-\beta E_k}$, and $H|\psi_j\rangle = E_j|\psi_j\rangle$. Inserting a complete set of harmonic oscillator stationary states $\{|n\rangle\}$, where $H_0|n\rangle = (n + 1/2)|n\rangle$, gives

$$\langle p | \psi_j \rangle = \sum_n \langle p | n \rangle \langle n | \psi_j \rangle \quad (2)$$

H can be constructed in harmonic oscillator basis by exploiting the property of ladder operators

$$a = \frac{q + ip}{\sqrt{2}}, \quad a^\dagger = \frac{q - ip}{\sqrt{2}} \quad (3)$$

$$a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (4)$$

Then we can construct the matrix element $\langle n | q | m \rangle$ and $\langle n | p | m \rangle$, and then construct $\langle n | H | m \rangle$ by doing various matrix multiplications from these two matrices. Once we have build $\langle n | H | m \rangle$, this can be diagonalised to obtain $\langle n | \psi_j \rangle$, and E_j and be plugged into Eq.(2) and w_j in Eq.(1). We now need $\langle p | n \rangle$, consider the matrix element $\langle p | p | n \rangle$

$$p\langle p | n \rangle = i\sqrt{\frac{n+1}{2}}\langle p | n+1 \rangle - i\sqrt{\frac{n}{2}}\langle p | n-1 \rangle \quad (5)$$

rearrange yields

$$\langle p | n+1 \rangle = \sqrt{\frac{n}{n+1}}\langle p | n-1 \rangle - ip\sqrt{\frac{2}{n+1}}\langle p | n \rangle \quad (6)$$

this is a recurrence relation, we just need to know $\langle p | 0 \rangle$ to deduce every other $\langle p | n \rangle$, using $a|0\rangle = 0$ and $\langle 0 | 0 \rangle = 1$ it is easy to show

$$\langle p | 0 \rangle = \pi^{-1/4} e^{-p^2/2} \quad (7)$$

then using $a|0\rangle = 0$ again in Eq.(6), we have $\langle p | 1 \rangle = -i\sqrt{2}p\langle p | 0 \rangle$, then plug $\langle p | 0 \rangle$ and $\langle p | 1 \rangle$ into Eq.(6) to obtain $\langle p | 2 \rangle$ etc. etc. Once we have $\langle p | n \rangle$, it can be plugged back into Eq.(2) to construct $\langle p | \psi_j \rangle$, which can then be plugged into Eq.(1) for $n(p)$.